

Week 1

This week we will:

- Have an overview on Business Finance - related to SLO 1
- Explore three different Financial Decisions - related to SLO 1
- Explore the Time Value of Money concept and its applications - related to SLO 2
- Learn about Timeline and how to identify correctly Present Value (PV), Future Value (FV), number of time periods (n) and interest rate per period (i) - related to SLO 2 & 3
- Learn about Simple interest - related to SLO 1

Learning Outcomes

At the end of this module you will:

SLO1:describe the principles and objectives of financial decision-making

SLO2:apply time-value of money techniques to the valuation of financial instruments and financial securities

SLO3:demonstrate an understanding of financial risk, its measurement, and its relationship to return

Goal of Business Finance

- Every finance manager or Chief Financial Officer (CFO) must aim to:
Maximise shareholders' wealth

- The goal is achieved by:
Analysing opportunities and selecting projects that provide the best cash flows

How to measure the goal?

- The growth of Share price and Dividend payment
Shareholders' wealth would be maxismised as a result
- Profit maximization is NOT the focus of Financial Management

What's Wrong With Profit?

- **Profit has no timeframe** – it is a snapshot in time

Do we want short-run or long-run profits?

- **Does not consider uncertainty of future**

What will income and expenses look like in the future?

- **Depends on accounting standards**

Accounting decisions such as depreciation choices can affect profits

- **Profit is primarily an accounting concept**

Profit does not take into account when funds are actually received

Profit does not pay bills, loans or dividends.....cash does!

Wealth Maximisation

- Cash flow is the key!
- Cash flow is a better measure of wealth
 - Cash flow (unlike profit) can pay bills, loans or dividends
- Finance measures the wealth created by business projects
 - Finance assesses project cash flows and when they occur
- Wealth maximization:
 - Wealth is maximised by projects that give the greatest net cash flow in today's dollars
 - Financial analysis allows for different projects to be compared and for a business to select the project that adds most to shareholder wealth

Market Value and Book Values

- **Accounting View**

In accounting we use **historical or book values** for assets

- **Finance View**

In finance we are **not** concerned with the past

Focus on **future cash flows** that are produced by assets

- Maximising shareholders' wealth = maximising the **market value** of shareholders' equity

Question

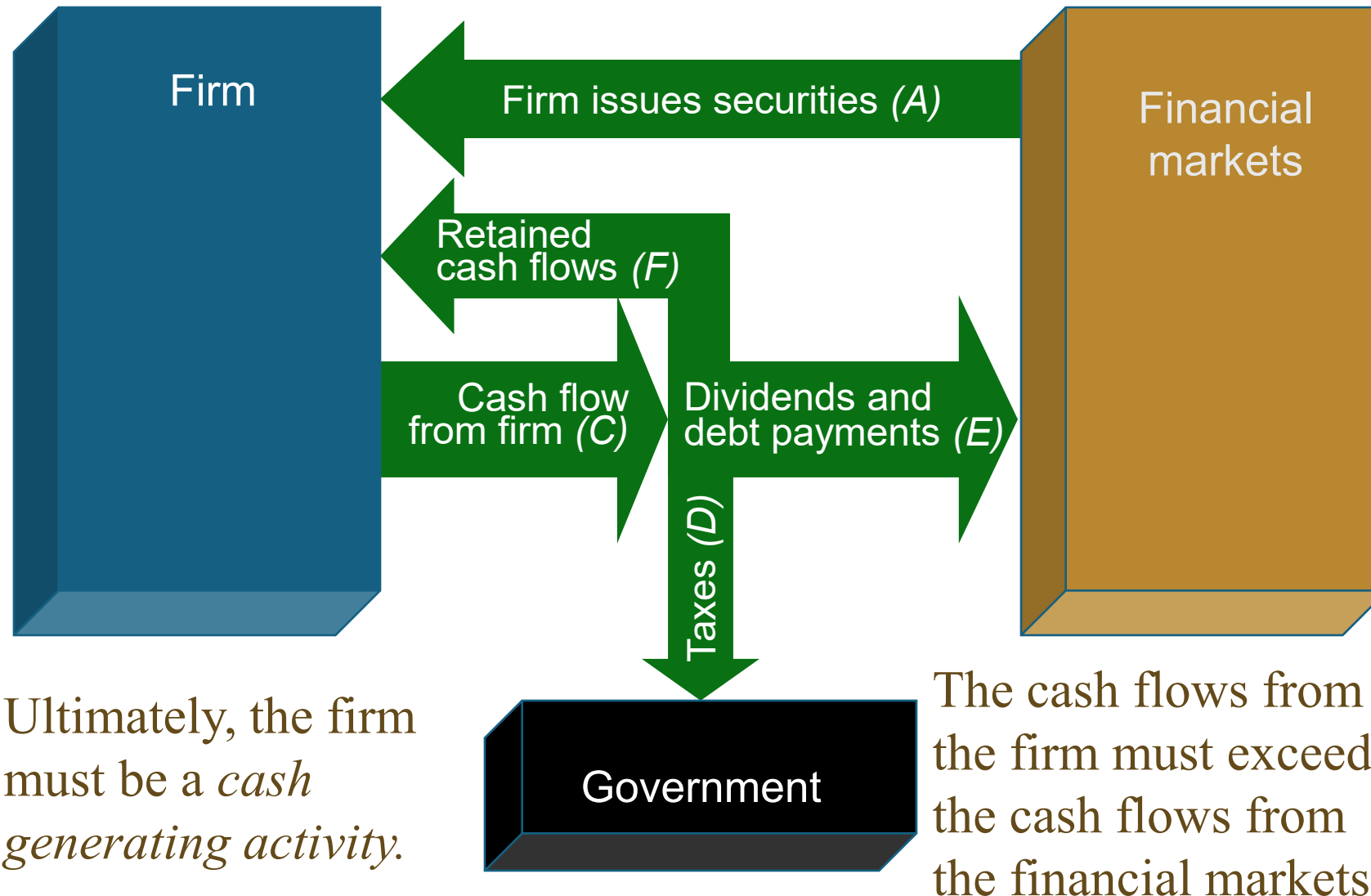
- **The wealth of shareholders is best measured by:**
 - a) The total assets of the firm
 - b) The total profits of the firm
 - c) The market value of the firm
 - d) The balance of Shareholders' funds of the firm

Answer: C

The Importance of Cash Flow

- If the firm is to prosper, it must:
 - Buy assets that generate more cash than they cost
 - Sell financial instruments that raise more cash than they cost
- The successful firm generates more cash than it uses

The Conceptual Flow of Cash



Cash Flow $\neq$ Accounting Income

- Do not confuse cash flow and accounting income
 - Non-Cash expense example: Depreciation
 - Non-Cash revenue example: Sales on Account

Cash Flow versus Accounting Income example

- The Midland Company refines and trades gold. At the end of the year, it sold 2,500 ounces of gold for \$1 million. The company had acquired the gold for \$900,000 at the beginning of the year. The company paid cash for the gold when it was purchased. Unfortunately it has yet to collect from the customer to whom the gold was sold.

Example continued

The midland company income statement
(Accounting View)
year ended December 31

Sales	\$1,000,000
-costs	<u>-900,000</u>
Profit	\$ 100,000

Example continued

The midland company income statement

(Financial view)

year ended December 31

Cash inflow	\$ 0
-Cash outflow	<u>-900,000</u>
net cash flow	- \$900,000

Conclusion: Cash Flow $\neq$ Accounting Income

Timing of cash flow

- The value of an investment made by a firm depends on the timing of cash flows.
- One dollar received today is worth more than one dollar received next year.

Timing of cash flow-example

- The Midland Company is attempting to choose between two proposals for new products. Both proposals will provide additional cash flows over a four-year period and will initially cost \$10,000. The cash flows from the proposals are as follows:

YEAR	NEW PRODUCT A	NEW PRODUCT B
1	\$ 0	\$4,000
2	0	4,000
3	0	4,000
4	<u>20,000</u>	<u>4,000</u>
Total	\$20,000	\$16,000

Timing of cash flow-example

Conclusion:

- At first it appears that new product A would be best.
- However, the cash flows from proposal B come earlier than those of A .
- Without more information, we cannot decide which set of cash flows would create the most value for the bondholders and shareholders.
- It depends on whether the value of getting cash from B up front outweighs the extra total cash from A .

Risk of cash flows

- The firm must consider risk.
- The amount and timing of cash flows are not usually known with certainty.
- Most investors have an aversion to risk.

Risk of cash flows-example

- The Midland Company is considering expanding operations overseas. It is evaluating Europe and Japan as possible sites. Europe is considered to be relatively safe, whereas operating in Japan is seen as very risky. In both cases the company would close down operations after one year.

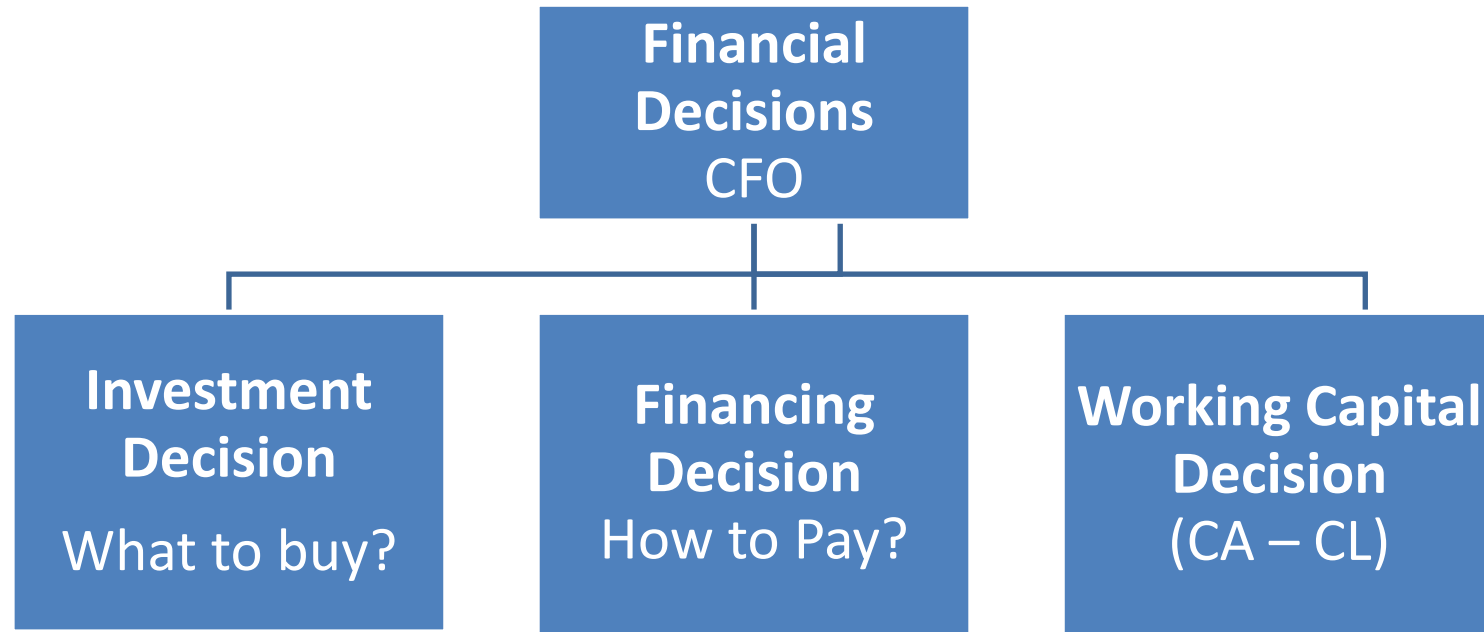
	PESSIMISTIC	MOST LIKELY	
	OPTIMISTIC		
Europe	\$75,000	\$100,000	\$125,000
Japan	0	150,000	200,000

Risk of cash flows-example

Conclusion:

- Japan appears to be riskier, but it also offers a higher expected level of cash flow.
- What is risk and how can it be defined?
- Corporate finance cannot avoid coping with risky alternatives, and much of our book is devoted to developing methods for evaluating risky opportunities.

Financial Management Decisions



Investment Decision

- Also called Capital Budgeting Decision
- How to determine the value of a long term asset....
 - Evaluate size, time and risk of future cash flows from the asset
 - Select the long term assets that create most shareholder wealth or
 - Composition of assets which will best facilitate the operations of the firm
 - The most important decision
 - Hard to reverse if wrong decision made
- We will be looking at ways of evaluating assets and calculating their impact on shareholder wealth

Financing Decision

- How to finance the project or investment...
- Determine the best source and mix of funds
- **Debt** (loan funds) which are borrowed and are a contractual claim on business assets
- **Equity** (owner's funds) which come from shareholders and are a residual claim on business assets
- Trade-off between risk and return
- Use of debt is called **gearing or leverage** and can improve return to shareholders
- **From an accounting perspective:**

Assets = Liabilities + Owner's Equity

(A = L + E)

Working Capital Decision

Managing short term assets and liabilities...

- Forms part of the investment decision
- Inventory management – How much inventory to hold?
- When to allow credit sales?
- When to pay suppliers?
- How much cash to hold?
- **Current Assets** – cash, liquid securities, inventory, accounts receivable
- **Current Liabilities** – accounts payable

- **Net Working Capital**
= Current Assets – Current Liabilities
(NWC = CA - CL)

Balance Sheet Model of the Firm

Total Value of Assets:

Current Assets

Fixed Assets

1 Tangible

2 Intangible

Total Firm Value to Investors:

Current
Liabilities

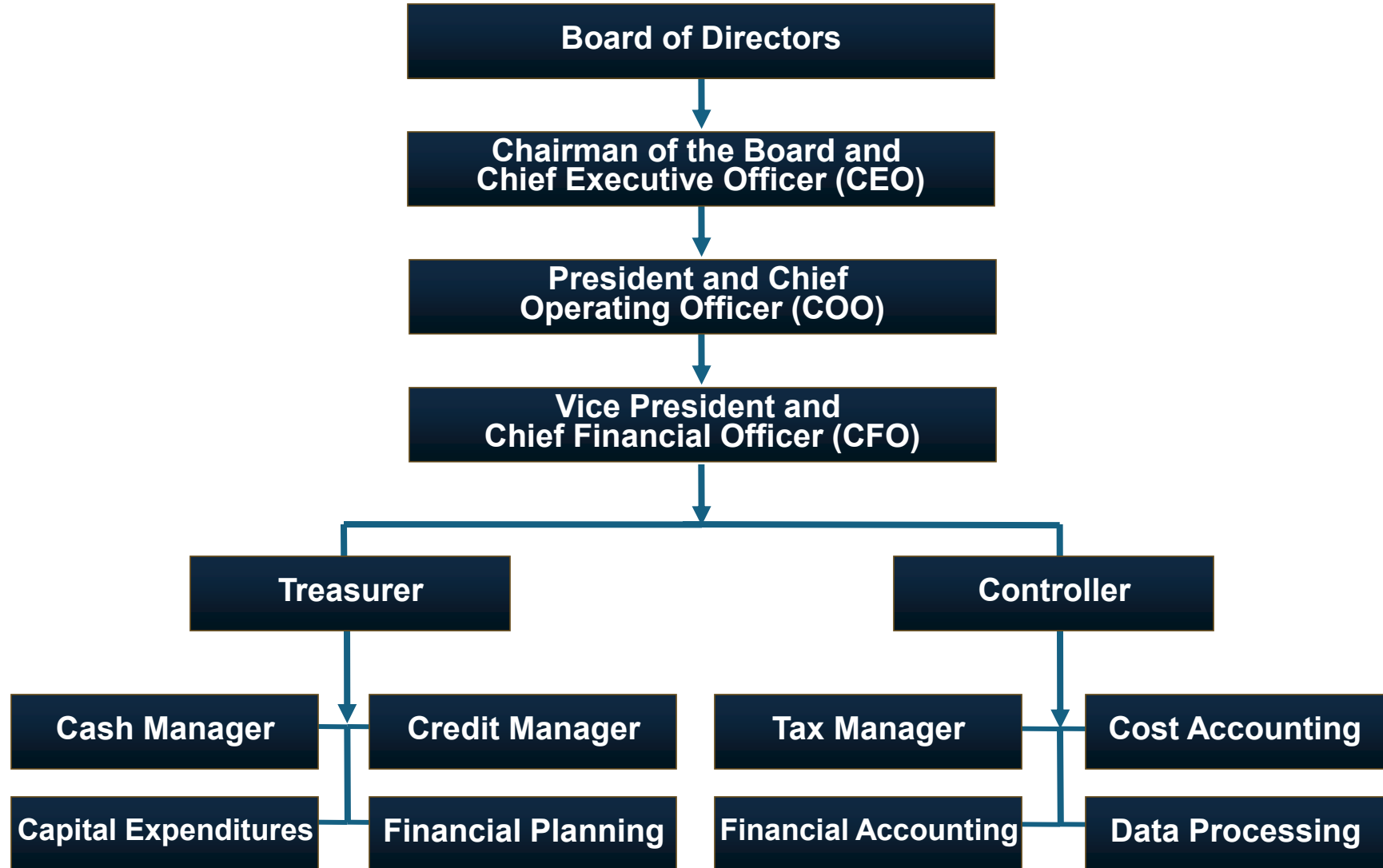
Long-Term
Debt

Shareholders'
Equity

The Financial Manager

- The firm's 3 main financial concerns are usually handled by a top officer and aides:
 - V.P. or Chief Financial Officer
 - Strategist, coordinator, authority
 - Treasurer
 - Cash flow, capital expenditure, capital structure
 - Controller
 - Accounting, information systems, taxes

Hypothetical Organization Chart



Question

• **Is the CFO is responsible for determining:**

- a) how suppliers will be paid
- b) the appropriate level of debt for a firm
- c) which projects a firm should undertake
- d) how to invest the firm's cash

Answer: ABCD

Forms of Business Organization

- The Sole Proprietorship
- The Partnership
 - General Partnership
 - Limited Partnership
- The Corporation



A Comparison

	Corporation	Partnership
Liquidity and marketability	Shares can be easily exchanged	Subject to substantial restrictions
Voting Rights	Usually each share gets one vote	General Partner is in charge; limited partners may have some voting rights
Taxation	Double	Partners pay taxes on distributions
Reinvestment and dividend payout	Broad latitude	All net cash flow is distributed to partners
Liability	Limited liability	General partners may have unlimited liability; limited partners enjoy limited liability
Continuity	Perpetual life	Limited life

The Agency Problem

- Agency relationship
 - Principal hires an agent to represent his/her interest
 - Stockholders (principals) hire managers (agents) to run the company
- Agency problem
 - Conflict of interest between principal and agent

Agency Cost

- Cost of Conflict of Interest
- Example:
 - Large investment positions firm for long term positive cash flow but has risk in short run
 - Owners want this investment – Increases firm value
 - Managers object – Risk may have personal cost
 - If managers prevail, foregone long term cash flow is the Agency Cost

Managerial Goals

- Managerial goals may be different from shareholder goals
 - Expensive perquisites
 - Survival
 - Independence
- Increased growth and size
 - Often lead to management reward
 - Not necessarily in best interest of shareholders

Managing Managers

- Managerial compensation
 - Incentives can be used to align management and stockholder interests
 - The incentives need to be structured carefully to make sure that they achieve their intended goal
- Corporate control
 - The threat of a takeover may result in better management
- Influence of other stakeholders

Time Value of Money (TVM) concept

- *Time Value of Money*
- Time value of money (TVM) suggests that a dollar today is worth **more** than a dollar in the future due to its future earning potential
- Wealth maximisation takes into account the time value of money
- **Goal:** Maximising shareholders' wealth = maximising the market value of the shareholders' equity in today's value

Time Value of Money (TVM) concept cont.

A person offers you \$500,000 for your sports car and want to **take the car today but pay for it 1 year later.**

Today



1 year



If you have to wait...

- You miss out on driving the car for a year
- You could sell the car for more in one year's time
- You may not get paid for it in one year

Why do You Want Payment Today?

- Why do you want payment for your car today?
- If you wait, you will want to be compensated for:
 - Sacrifice (not being able to drive it)
 - Inflation (change in prices over time)
 - Risk (may not get paid)
- For you to choose to receive payment later you want to be paid for the above factors. **Interest** represents all these factors.

$$\text{Future Value} = \text{Present Value} + \text{Interest}$$

How much **interest** is enough ? Based on the level of **risk** taken

Risk-Return Trade Off

If you had \$100,000 to lend or invest, who would you give it to?



Banks

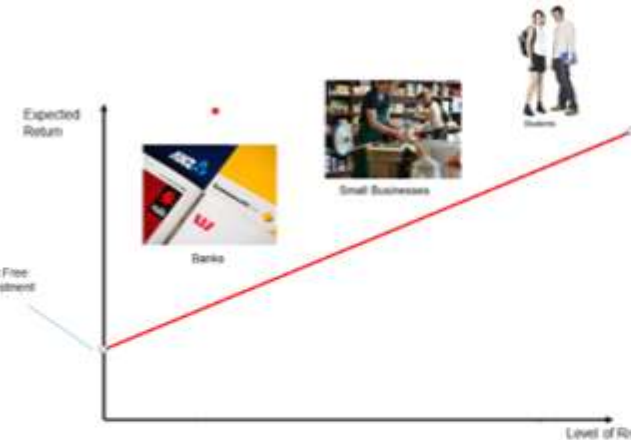


Students



Small Businesses

What return would you want from each group? Why?



Risk free return is the theoretical return from an investment with zero risk eg government bond

Risk

The actual outcome is different to what was expected

As risk increases investors require higher returns

If risk is taken then it must be rewarded

In this case:

- Lowest interest is expected if lending to Banks (lowest risk), followed by Small Businesses and Students

Question

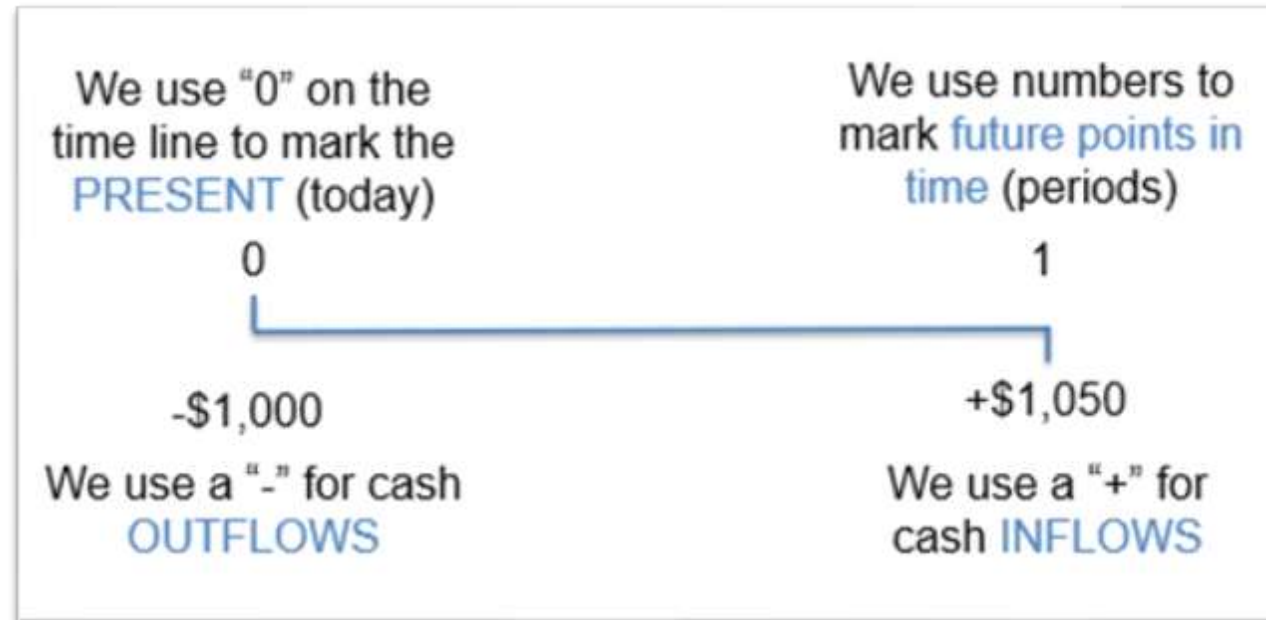
- **The trade-off between expected return and risk has a:**
 - a)horizontal slope
 - b)downward slope
 - c)vertical slope
 - d)upward slope

Answer: D

Timeline: Present Value and Future Value

- **Time Lines**
- A line or chart showing a company's cash inflows and outflows
- Can also be used to provide information about the cash flow
- Time lines can show simple or more complex cash flows
- Each point on the line represents a point in time

- Time Lines offer a **clear picture** of when and where cash flows occur and **Business Finance is ALL ABOUT CASH FLOWS**



Task

- ***Present Value & Future Value***

- Present Value (PV) is usually the value today (time point “0”)
- Future value (FV) is the value at some point in the future
- Draw a time line in the box below and mark the following time points
 - Today
 - The year you were born
 - The year you will graduate from UTS

Timeline: Number of time periods (n) and Interest rate period (i)

- **Time Periods (n)**
- **Time Periods** are the “gap” between points
- Time periods are usually measured in years but may also be **months, days, quarters or half years**
- The difference in the measurement of a time period will result in different time periods over the same time length
- For example:
- Let's consider a 4 year timeline below:
 - If we consider/define each year is 1 time period, in total, we will have 4 time periods



- The time line above will be different for different time periods:
- Half-Years or Semi-annuals: We consider each half-year is 1 time period. That means, each year will have 2 time periods. Over the same 4 year timeline, there will be in total 8 time periods (2×4)



- Similarly we have:
- Quarters: 16 time periods 3 months apart
- Months: 48 time periods 1 month apart
- Days: 1,460 time periods 1 day apart (always consider 365 days in 1 financial year)

Annual Percentage Rate and Interest rate per period (i)

- Annual Percentage Rate: The Interest Rate that you will earn in **1 YEAR**. Normally indicated as per annum, p.a or per year
- Interest Rate per period (i): The interest rate that is earned **PER PERIOD**
- Interest Rate per period (i) =
$$\frac{\text{Annual percentage Rate}}{\text{The number of periods per 1 year}}$$
- For example: An investment is earning 12% interest **per annum**, in which **every half year is considered a period**.
- That means:
 - The **annual** percentage rate is 12%
 - The interest rate **per period** (i) is $12\%/2 = 6\%$
- The variable “interest rate per period” (i) is used frequently in **MOST OF FINANCE CALCULATIONS**

Questions

- You can earn 12% Interest per annum by investing with MYBank. Calculate **n (number of time periods)** and **i (interest rate PER PERIOD)**
- 3 years with interest paid half yearly (semi annually)
- 1.5 years with interest paid monthly
- 182 days with interest paid yearly

Answer

"3 years with interest paid half yearly (semi annually)"

- Time length: **3 years**
- "Paid half yearly" -> **Every half year** is considered as **1 period**.
- There will be **2 PERIODS PER YEAR**
- 12 % interest **PER YEAR**

Put all together:

- Number of time periods (n): $n = 3 * 2 = 6$ time periods
- Interest per period (i): $i = 12\% / 2 = 6\%$ per period

Answer

"1.5 years with interest paid monthly"

- Time length: **1.5 years**
- "Paid monthly" -> **Every month** is considered as **1 period**.
- There will be **12 PERIODS PER YEAR**
- 12 % interest **PER YEAR**

Put all together:

- Number of time periods (n): $n = 1.5 * 12 = 18$ time periods
- Interest per period (i): $i = 12\%/12 = 1\%$ per period

Answer

"182 days with interest paid yearly"

- Time length: **182 days**
- "Paid yearly" -> **Every year** is considered as **1 period**.
- There will be **less than 1 PERIOD (365 days in a period but the time length is only 182 days)**
- 12 % interest **PER YEAR**

Put all together:

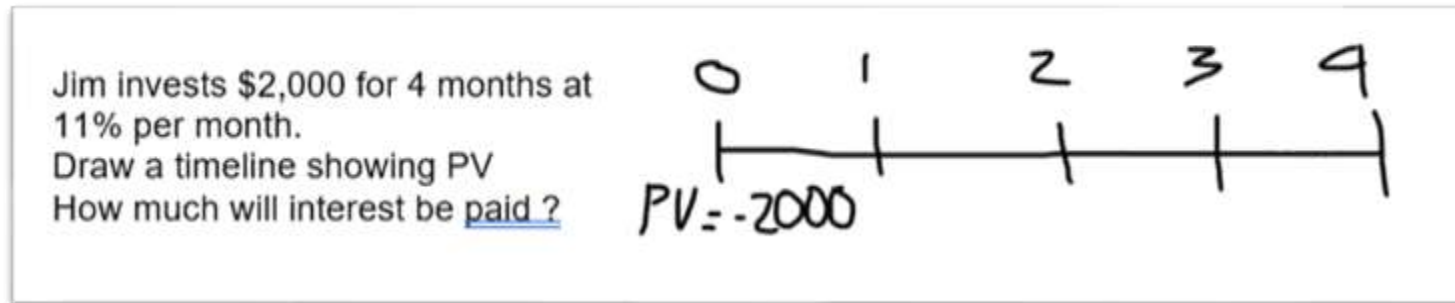
- Number of time period (n): $n = 182/365$ (Remember, it's assumed that we always have 365 days in a year)

Simple Interest

- Simple Interest is a quick method of calculating the interest **charged on a loan** or the interest **earned on a deposit**
- Simple Interest is calculated on the **original principal only**
- There is **no** interest earned on interest
- Used in the valuation of short term money market securities where the time period is less than 12 months (Commercial Bills)

Calculating Simple Interest

- Simple Interest is calculated on the **original principal only**
 - Interest = Present Value x Interest rate x Time Periods



$$\text{INT} = \text{PV} \times i \times n$$

- Interest earned over 4 months = $\text{PV} \times i \times n = 2000 \times 11\% \times 4 = \880

Calculating Future value

- **Future Value** is the original principal investment (PV) plus the amount of simple interest (**INT**)

Future Value = Present Value + Simple Interest

$$\mathbf{FV = PV + INT}$$

$$\mathbf{FV = PV + (PV \times i \times n)}$$

$$\mathbf{FV = PV(1 + i \times n)}$$

Jim invests \$2,000 for 4 months at 11% per month.
How much will he have in 4 months' time?

- Amount of money in 4 months time is: $FV = PV (1 + i * n) = 2000 * (1 + 11\% * 4) = 2880$

- If we have the Future Value (FV) but need the Present Value (PV) we can rearrange the formula to calculate PV

$$\mathbf{FV = PV(1+i \times n)}$$

Rearrange to solve for PV.....

$$\mathbf{PV = FV/(1 + i \times n)}$$

Jim accumulated an amount of \$2,880 by investing for 4 months at a simple interest rate of 11% per month.

How much was his original investment?

Original investment was: $PV = FV/(1+i*n) = 2880/(1+11\%*4) = 2000$

Question 1

- You invest \$80,000 for 180 days and earn 8% pa Simple Interest.
 - Draw a timeline showing cash inflows (+) and outflows (-)
 - How much money will you have after 180 days?
 - How much money will you have after 70 days?
 - What is the relationship between i and FV?
 - What is the relationship between n and FV?

Question 1

- You invest \$80,000 for 180 days and earn 8% pa Simple Interest.
 - How much money will you have after 180 days?
 - $FV = PV(1+i \times n)$
 - $FV = 80,000 * (1 + 8\% * 180/365)$
 - $FV = 83,156.16$
 - How much money will you have after 70 days?
 - $FV = PV(1+i \times n)$
 - $FV = 80,000 * (1 + 8\% * 70/365)$
 - $FV = 81,227.40$
 - What is the relationship between i and FV ?
 - Higher i , higher FV
 - What is the relationship between n and FV ?
 - Higher n , higher FV

Question 2

- You can invest 90 days and earn 7% pa Simple Interest. If you need to accumulate \$100,000 how much should you invest today? Draw a timeline.

Question 2

You can invest 90 days and earn 7% pa Simple Interest. If you need to accumulate \$100,000 how much should you invest today? Draw a timeline.



$$PV = FV / (1 + i \times n)$$

$$PV = 100,000 / (1 + 7\% \times 90/365)$$

$$PV = 98,303.26$$